Unit 202 — Visual Production Proof Review

Package: CC-11.8 · Generated: 2026-08-24T21:04:50.062Z

TWO-ASSET AUTOMATED PRODUCTION PIPELINE PROOF — FOR INDEPENDENT REVIEW BEFORE ANY WIDER PRODUCTION RUN

This document proves the automated visual-production pipeline (Claude orchestrates, Gemini renders, Claude audits) end-to-end on two real assets. It is not a bulk production run and does not constitute Product Owner approval of either candidate image.

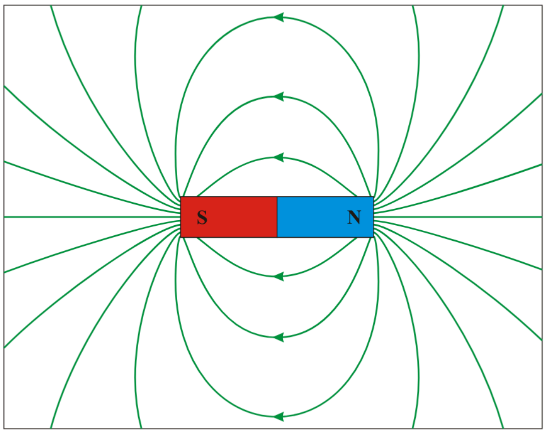
Assets attempted	2
Passed (first attempt)	1
Retry-passed	0
Human review required	1
Failed / incomplete	0
Total Gemini calls made	3 (of 4 maximum permitted)

Bar magnet and magnetic field

unit202.magnet.field

Curriculum context	LO5 — lesson.magnetism.fundamentals
Pedagogical role	PHENOMENON
Production class	HYBRID
Purpose	Show a bar magnet with its external magnetic field pattern, N to S, as the basis for flux/flux-density teaching.

REFERENCE



Wikimedia Commons — DipolMagnet.svg — https://commons.wikimedia.org/wiki/File:DipolMagnet.svg

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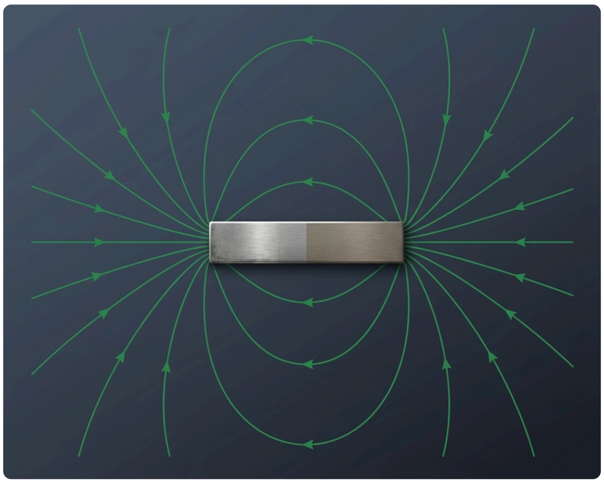
FINAL ALP ARTWORK (ATTEMPT 1)



Model: gemini-3.1-flash-image · Generated: 2026-08-24T20:56:55.079Z

AUDIT VERDICT: PASS (retry count: 0)

Immutable fact	Result	Note
meaningful field-line geometry	PASS	Concentric-loop dipole field pattern with the same tiered arc structure (multiple nested loops above and below the bar plus one straight central line each side) as the DipolMagnet.svg reference. Topology matches: loops emerge from one pole, curve around, and terminate at the other, symmetric above/below the bar.
external field direction runs N to S	PASS	Field-line circulation is self-consistent across every visible loop (all arrowheads follow the same rotational sense, exiting one pole and entering the other on both the top and bottom loop sets) -- no arrow contradicts its neighbours. The reference's N/S colour-coding (red/blue halves) and letter labels were correctly NOT baked into the image per this proof's clean-base-art override -- pole identity (which end is N vs S) is therefore not independently re-verifiable from this image alone and is deferred to ALP's deterministic N/S label overlay, exactly as the asset's own deterministicOverlayResponsibilities design intends. This is the expected division of responsibility, not a generation defect.
field-line density used meaningfully where density is taught	PASS	Not applicable to this state (density_comparison=false) -- no density-comparison claim is made by this image, correctly.

Prohibited change	Result	Note
do not draw field lines that reverse direction or cross incorrectly	PASS	All loops are cleanly nested with no line-crossing artefacts, and no arrowhead reverses direction against its own loop.

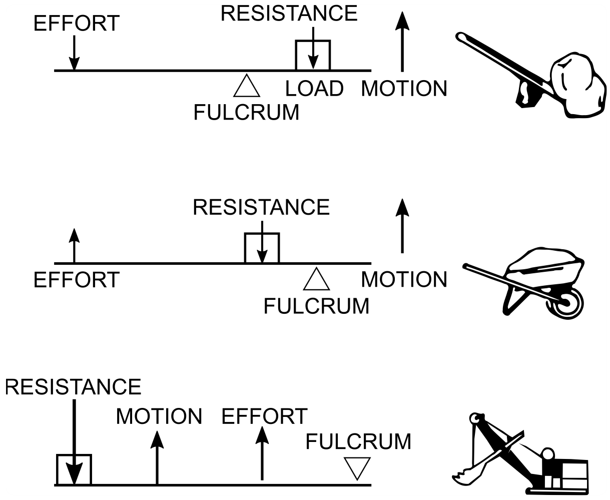
Baked labels found	No -- clean base art confirmed
Style compliance	Background is a muted dark slate/blue-grey gradient, a very close match to the governed #151821 -> #262B38 specification -- no black void, no neon, no scenery. The magnet bar is rendered as brushed metal with real material/lighting cues (premium, tactile, industrial-precision character, not cartoon or flat-icon). Field-line colour (green) happens to match the pre-existing 'magnetic field = green/turquoise' semantic-colour convention referenced in master-prompt.ts, but the new style guide document does not yet explicitly state semantic field colours -- noted as a prompt-builder gap for a future package, not a defect in this image (the reference itself uses green, and the result is consistent with the platform's existing convention).
Physical plausibility	None found -- a solid metal bar with a coherent surrounding field halo is physically plausible and internally consistent.
Overall findings	Strong pass. Topologically faithful redraw of the reference's dipole field-line geometry in premium ALP style, correctly omitting baked labels/pole colour-coding per this proof's clean-base-art policy, with a governed-style-compliant background. No immutable fact violated, no prohibited change made. Recommended: PASS, no retry needed.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.magnet.field\unit202.magnet.field-master-v1.png` (sha256 8033c45b43442922...)
Derivative path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.magnet.field\unit202.magnet.field-derivative-v1.png`
Reference SHA-256	`577d68f467b6cd28c4452026673eed7e9ba34ac2d0389a6d4fece2131764375d`

Lever — Class I

unit202.levers.class-1

Curriculum context	LO3 — lesson.foundation.physics.simple-machines
Pedagogical role	CONFIGURATION
Production class	HYBRID
Purpose	Show a Class I lever arrangement (fulcrum between effort and load) so a learner can recognise it from the arrangement itself.

REFERENCE



Pearson Scott Foresman — Lever (PSF).svg — https://commons.wikimedia.org/wiki/File:Lever_(PSF).svg

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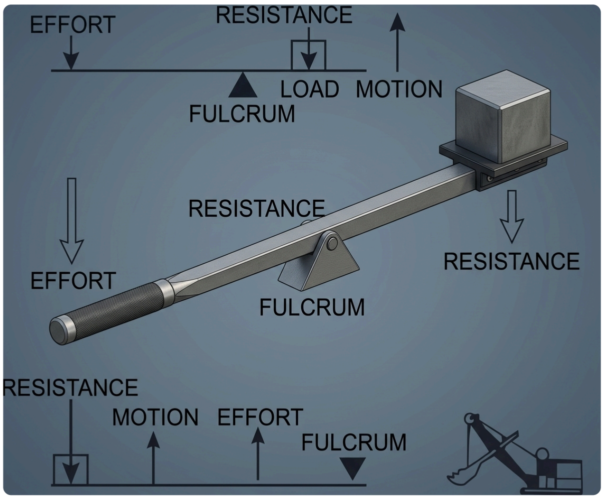
FINAL ALP ARTWORK (ATTEMPT 2)



Model: gemini-3.1-flash-image · Generated: 2026-08-24T21:03:56.874Z

AUDIT VERDICT: HUMAN_REVIEW_REQUIRED (retry count: 1)

Immutable fact	Result	Note
fulcrum positioned between the effort point and the load point	UNCERTAIN	In the central 3D render itself, the fulcrum wedge IS positioned between the effort handle (far left) and the load cube (far right), which is correct in isolation. But the corrected prompt's two explicit instructions -- (a) render exactly one lever diagram, and (b) bake no text -- were both still not followed: the top and bottom schematic reproductions from the source composite are still present in full, including the bottom schematic where FULCRUM sits past EFFORT at the far right end (a Class II/III-type ordering), so the delivered image still juxtaposes a non-Class-I arrangement against the Class I asset.

Prohibited change	Result	Note
do not blend this with the Class II or Class III arrangement	FAIL	Unchanged from attempt 1: the bottom schematic (order RESISTANCE, MOTION, EFFORT, FULCRUM left-to-right) still places the fulcrum at the far end past the effort point, which is not a Class I ordering, and still appears in the same delivered image.

Baked labels found	YES -- style-guide violation
Style compliance	Background and central 3D lever-rig render remain a strong style match. Unchanged from attempt 1 in every respect the correction targeted.
Physical plausibility	None beyond the composite-diagram and labelling defects.
Overall findings	The bounded correction attempt did not resolve either defect identified in attempt 1: the model again reproduced the full multi-diagram composite reference end-to-end (including the non-Class-I bottom schematic) and again baked every text label

	onto the image, despite the correction prompt explicitly and separately instructing against both. This exhausts this asset's permitted attempt budget (1 initial + 1 correction, task brief E7). Per the brief's explicit instruction not to burn further attempts chasing a fix, this asset is marked HUMAN_REVIEW_REQUIRED rather than retried again. Likely root cause for a future prompt-engineering pass: the reference image itself is a dense multi-diagram composite, and the model appears to be treating the entire reference frame as content to reproduce rather than as a geometry source to selectively extract from -- a cropped/pre-isolated reference (showing only the Class I portion) would likely be a more reliable fix than prompt wording alone.
Master path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.levers.class-1\unit202.levers.class-1-master-v2.png` (sha256 e903c417179c5e60...)
Derivative path	`D:\Development\adaptive-learning-platform\reports\instructional-visuals\premium-artwork\proof\unit202.levers.class-1\unit202.levers.class-1-derivative-v2.png`
Reference SHA-256	`e3a8bf457be32da9dc950b0991446884638cdf00f7807cbd4ae0583fee2b4450`